

Next-Track Session Recommendation: Experiment Synthesis

July 15, 2026

Abstract

This document is the living synthesis of the *next-track* sequence recommendation campaigns; the markdown reports in **experiments/** (filenames containing **nexttrack**) are the primary record. The task is next-item *retrieval*: given a listening *session* (an ordered sequence of tracks), rank the catalog and recover the next distinct track the user actually played, scored on a single leak-free chronological *next-distinct* split (7154 sessions / 19 402 items; 5723 train / 1431 test; cold-item rate 0.55). The headline metric is Recall@10, reported beside mean reciprocal rank (MRR) and the graded-relevance diagnostics artist@10 / genre@10 / artist_mrr. Seven campaigns are consolidated. A session GRU alone only *ties* the first-order Markov co-listening bar (R@10 0.1069) on Recall@10 and loses on MRR, and neither loss-tuning nor any network topology (depth, width, residual, direction) beats it. The breakthrough is *rank fusion*: an R+M z-blend (GRU $\times$ Markov) beats the bar by $\sim +61\%$ Recall@10 (0.1726, three-seed confirmed), because the two legs rank *different* correct tracks; a learned XGB stacker underperforms the fixed z-blend and adding redundant neural legs does not help. The one genuinely additive new signal is *content*: an item-representation sweep found latent *dimension* (not the compressor) to be the lever, with a peak at PCA-192, and on that rich space a three-leg R+M+C z-blend (GRU $\times$ Markov $\times$ content-kNN) is the current best-on-record row at Recall@10 0.1859, MRR 0.1025 (paired Δ over the prior champion $+0.0133$, CI above zero). Two caveats bound the crown — the reproduction confirm had zero spread because the trainer is deterministic (it verified determinism, not seed-robustness), and the paired- Δ lower bound clears zero by only $+0.0007$ on a single split. All ranking predictors here are train-only and off-server, so a confirmed win is a best-on-record *row*, not a promoted model, and the prior champion R+M on AE-64 remains the productization anchor.

1 Introduction

This document is the living synthesis of the next-track sequence-recommendation campaigns; the markdown reports in **experiments/** are the primary record. Each section transcribes (never re-computes) the quantitative results of a campaign, preserves its run and dataset identifiers verbatim for traceability, and adds the rationale and interpretation the terse reports omit. This instance was cloned from a sibling taste-fit / rotation lab and retargeted to a sequence ranking goal; the rotation campaigns are a separate line of work and are *not* covered here.

Task and split. The lab predicts, within a listening *session*, which track the user plays *next* — a next-item retrieval task. The model consumes the ordered session prefix and scores every catalog item; candidates are ranked and the held-out next *distinct* track is the target (a leave-last-out evaluation that excludes prefix items, matching the Markov parity convention). All work runs on one sequence artifact, **seq-20260713-014226** (7154 sessions / 19 402 items), split *chronologically by session start* into 5723 train / 1431 test at cut 2024-08-24. The defining property of this split is its

cold-item rate of 0.55: more than half of test-session target tracks never appear in any training session. This is the leak-free forward reality, and it caps absolute recall by construction — every result is read against the incumbent baseline bar, never an absolute number.

Metrics. The headline metric is **Recall@10** (R@10, the fraction of test sessions whose held-out next track appears in the top ten retrieved), reported beside **mean reciprocal rank** (MRR, the average inverse rank of the true next track) and Recall@20. Since 2026-07-14 the evaluation also emits first-class *graded-relevance* metrics over the same prefix-excluded ranking (Section 7): **artist@k** (a top-*k* candidate shares the true next track’s artist), **genre@k** (likewise on primary genre), and **artist_mrr** (MRR of the first same-artist candidate). All confidence intervals are bootstrap 95% CIs over 2000 resamples of the 1431 test sessions, and the decisive statistic is the *paired* per-session Δ against a named reference. A board *win* requires beating the reference on *both* R@10 and MRR beyond non-overlapping marginal CIs (the strict bar), then a reproduction confirm; the paired- Δ CI-entirely-above-zero test is the more powerful alternative statistic (it crowned both champions) and can disagree with the strict bar. The graded metrics are the honest scorecard: exact recall is modest by construction, but the champion lands the right *artist* in the top ten $\sim 32\%$ of the time and the right *genre* $\sim 40\%$ — far more useful than the $\sim 17\%$ exact number the cold-item rate imposes.

The standing champion and the productization anchor. The current best-on-record *row* is a three-leg rank-fusion blend, *R+M+C* (a session GRU $\times$ a first-order Markov co-occurrence model $\times$ a content-kNN leg) on a PCA-192 item space, at R@10 0.1859, MRR 0.1025 (Section 10). The prior champion, the two-leg *R+M* z-blend on the 64-dim song-autoencoder space (AE-64), at R@10 0.1726 / MRR 0.0963 (Section 6), remains the *productization anchor*: it is the two-leg blend that maps directly onto the sibling app’s transition slot, and it stays the deployable model until the three-leg blend is registered as a predictor. **No ranking predictor here is best-models-promotable:** they are train-only (no `predict` subcommand), and *R+M+C* is an off-server fusion driver, so a confirmed win is a best-on-record row, not a promoted model.

2 Background: the sequence task, the baselines, and the metrics

This section gives the minimum theory for a reader who knows machine learning but not this repo. It is keyed to the project’s living facts file and grows only when a new predictor family or metric enters the reports.

The models and baselines in play.

Session GRU / LSTM (`seq-nexttrack`). A recurrent neural network that reads the ordered session prefix one item at a time (each item represented by its item-latent vector) and emits a predicted next-item latent; candidates are ranked by cosine similarity to that prediction. The GRU (gated recurrent unit) and LSTM (long short-term memory) are the two recurrent cells tried. The recurrence captures *session context* — the trajectory of the listening session — rather than a single previous item.

Training loss: InfoNCE versus cosine. The sequence model can be trained to regress the next-item latent by *cosine* loss (pull the prediction toward the true next latent) or by **InfoNCE**, a contrastive objective that pulls the prediction toward the true next item and

pushes it away from a batch of negatives. InfoNCE is the single most important lever in the whole arc: it is $\approx 2.3\times$ cosine on Recall@10, because ranking retrieval rewards a representation that *separates* the true next item from distractors, which a contrastive loss optimizes directly and a pointwise regression does not.

First-order Markov co-occurrence (the bar, B2). A train-only transition model: a `track_bigram` counting “what track follows what” with artist/genre back-off for unseen pairs (the transition term the sibling app already ships). It directly *memorizes* adjacency, which is exactly what exact next-track ranking rewards, and it is the reference bar every arm is measured against. A second-order Markov (trigram) back-off was also tried and is essentially neutral — sessions are too short/sparse for a trigram to fire, so it falls through to the bigram.

item2vec. A skip-gram (SGNS) co-listening embedding trained on the *train* sessions only (a leak-safe analogue of word2vec, where “words” are tracks and “sentences” are sessions); candidates are ranked by cosine to the mean-pooled prefix. On 5723 train sessions it is too sparse to localize and is the weakest leg on record; the full-history co-listening vectors are deliberately *not* used because they leak across the split.

content-kNN. A training-free retrieval leg: cosine of each candidate’s item latent to the prefix latents (max-aggregated). It is weak standalone on exact recall but captures *vibe* similarity the counts miss (its genre@10 already edges the Markov bar), and it is the one genuinely additive third leg for the fusion blend.

Rank-fusion z-blend versus a learned stacker. Two ways to combine legs. The **z-blend** normalizes each leg’s candidate scores (z-score) and takes a weighted sum ($\alpha \cdot z(\text{leg}) + \dots$) — a fixed, parameter-light fuser. The **XGB stacker** is a gradient-boosted meta-learner over per-candidate features (leg scores + popularity + recency + artist/genre-match), trained pointwise with sampled negatives. Across two campaigns the fixed z-blend *beats* the learned stacker decisively, because the stacker’s pointwise binary training does not optimize the ranking objective as well as directly fusing normalized scores.

Item-latent spaces: song-AE versus PCA, and the wrong-collection trap. The sequence model predicts *into*, and retrieves *over*, a per-item latent space, so the choice of that space is a first-class axis. Three families appear:

AE-64 (`spotify_tracks_song_ae`). A 64-dim nonlinear *song-autoencoder* latent, the original champion item space. An autoencoder compresses each item by reconstructing it through a bottleneck; its equal-weight-per-block reconstruction loss spreads capacity evenly across content, acoustic and catalog blocks.

PCA- k (`spotify_tracks_song_pca $k$`). A *linear* principal components compression at dim $k \in \{32, 64, 128, 192, 256\}$, which orders dimensions by explained variance and decorrelates them. As Sections 8–9 show, this variance-ordered, decorrelated geometry is *easier* for the GRU to regress into than the AE’s entangled equal-weight latents, so PCA beats the AE at every dim ≥ 64 .

The wrong-collection trap. Both of the above are fit *only* from the leak-safe `spotify_tracks_content` source (intrinsic content/acoustic/catalog fields; no behavioral or target field). The default `spotify_tracks` 200-dim collection is a *behavioral* co-listening embedding that soft-leaks replay behavior and must **never** be used as an item space.

3 Next-track leaderboard

Table 1 is the standing best-on-record board. It is on the single leak-free next-distinct split and sorted by Recall@10. **Every ranking predictor here is train-only and off-server**, so the “champion” is a best-on-record row, not a promoted, app-exportable model; the prior champion R+M on AE-64 remains the productization anchor (Section 1). The item spaces are all fit from the leak-safe content source (Section 2).

Table 1: Next-track best-on-record board, on the leak-free next-distinct split (1431 test sessions; bootstrap 95% CI over 2000 resamples). R@10, MRR, artist@10, genre@10 higher-better; the first-order Markov model is the incumbent reference bar. Champion row and prior champion bolded; best cell per metric bolded.

Model	R@10 [95% CI]	MRR	art@10	gen@10	Note
R+M+C, PCA-192 (champion row)	0.1859 [0.166, 0.207]	0.1025	0.331	0.394	best on record
R+M+C, PCA-128	0.1824 [0.162, 0.203]	0.0993	0.332	0.396	content additive
R+M, AE-64 (prior champion)	0.1726 [0.154, 0.192]	0.0963	0.322	0.397	prod. anchor; 3-seed
R+M+C, AE-64 (graded-preferred)	0.1740 [0.154, 0.194]	0.1014	0.330	0.400	content only tied
R+M, PCA-192	0.1712 [0.151, 0.192]	0.0952	0.329	0.398	$\approx$ prior champion
R+M, PCA-128	0.1670 [0.148, 0.187]	0.0920	0.329	0.398	GRU gain not carried
R+A+M, AE-64	0.1684	0.0943	—	—	ANN leg redundant
LSTM single, PCA-128	0.1272 [0.110, 0.145]	0.0495	0.335	0.458	best single; $\ll$ blend
GRU single, PCA-192	0.1230 [0.106, 0.140]	0.0534	0.338	0.450	dim point-optimum
GRU single, PCA-128 (base)	0.1146 [0.099, 0.132]	0.0489	0.333	0.452	confirmed base space
GRU single, PCA-256	0.1167 [0.101, 0.134]	0.0502	0.338	0.458	past the dim peak
B2 first-order Markov (bar)	0.1069 [0.092, 0.124]	0.0694	0.298	0.389	app incumbent
GRU single, AE-64 (control)	0.0957 [0.081, 0.111]	0.0411	0.294	0.450	bake-off control
ANN pooled-MLP, PCA-128	0.0901 [0.076, 0.105]	0.0394	0.299	0.435	loses to the GRU
XGB stacker (M+G / M+G+A)	0.080–0.087	0.042–0.049	0.31	0.38	combiner underperforms
content-kNN (song-AE)	0.0636 [0.050, 0.077]	0.0321	0.272	0.405	the additive C leg
item2vec (train-only SGNS)	0.0517	0.0214	0.166	0.312	drags the blend
B0 popularity / B1 recency	0.0007	—	—	—	floors

4 The sequence sweep: a GRU-infonce ties the Markov bar

Rationale. The instance’s opening campaign asked the founding question: does a session-context sequence model — a GRU/LSTM over the ordered prefix, predicting the next track’s 64-dim song-AE latent and retrieving by cosine — beat the first-order Markov transition model the sibling app already ships, *forward*, on the leak-free session split? Four arms crossed the recurrent cell (GRU, LSTM), capacity (**hidden** 128, 256) and, decisively, the training loss (cosine versus InfoNCE), at 60 epochs.

Results. No arm beat the bar. The best arm (M3, GRU **hidden**128 InfoNCE) reached a statistical *tie* with the Markov bar on R@10 (0.0894, paired- Δ CI includes zero) but *lost* on MRR (0.0388 versus 0.0694) and Recall@20. The dominant lever was the loss objective: InfoNCE was $\approx 2.3\times$ cosine on R@10 (0.0894 versus 0.0391), while capacity (**h256**) and the LSTM cell barely moved the cosine arms (all ~ 0.038 – 0.041). The popularity and recency floors sat near zero (R@10 0.0007).

Interpretation. A latent-plus-single-cosine-hop is a harder route to the *exact* next track than a co-listening bigram: the bigram directly memorizes “what follows what,” while the GRU must land near the right point in a 64-d content space and then retrieve, losing on precise ranking (MRR). The

finding is consistent with a taste-drift reality — 0.55 cold items, identity/adjacency-driven behavior — and it set the agenda: InfoNCE is the base loss, and the natural next arm is a *blend* of the GRU’s candidate scores with the Markov bigram. *Source: 2026-07-13-nexttrack-sequence-sweep.md.*

5 Topology scan: depth, width and direction are a spent lever

Rationale. With the loss fixed at InfoNCE, this scan isolated network *topology*: does depth (stacked layers), width (`hidden` 256), an inter-layer residual stack or bidirectionality lift the sequence model over the Markov bar? Because a bidirectional RNN under per-step teacher forcing would leak future items, the bidirectional arm used a leak-free last-item objective (encode the full prefix, predict only the held-out next item).

Results. Refuted. No arm beat the bar on both metrics. The best arm (width `h256`, R@10 0.0957) is a *tie* on R@10 and loses on MRR (0.0411 versus 0.0694). Depth is flat 1→2 layers (0.0894→0.0922, in-band) and *hurts* at 3 layers (0.0825, out-of-band below the bar); the residual stack over depth-2/h256 added nothing. The bidirectional arm collapsed (0.0482), but this is an *objective* artifact, not a directionality verdict: the leak-free last-item objective supplies ~ 1 training target per session versus teacher forcing’s ~ 13 next-step pairs, so it trained on $\sim 13\times$ less signal.

Interpretation. Architecture is not the lever for next-track on this corpus. The transferable latent-retrieval signal caps at R@10 ~ 0.09 – 0.10 across depth, width, residual and direction, below the co-listening bigram — consistent with the whole arc that the loss objective moved the needle and capacity/topology do not. The one remaining unexplored lever was the candidate framing / fusion, which the next campaign pulled decisively. *Source: 2026-07-13-nexttrack-topology-scan.md.*

6 The GRU×Markov fusion champion, and why more ensembling fails

Rationale. The sweep and the topology scan agreed that the best single GRU only *ties* the Markov bar on R@10 while losing on MRR. This campaign asked whether *combining* models beats the bar, and how far ensembling goes. Three legs were defined, each emitting full-vocab candidate scores: **M** = the train-only Markov bigram (the bar), **R** = the GRU-infonce-h256 recurrence, and **A** = an ANN bag-of-prefix MLP (a feed-forward twin of R). They were combined by a weighted z-blend and, as a learned alternative, an XGB stacker.

Results. The **R+M z-blend** ($\alpha=0.5$) *beats the bar decisively*: R@10 0.1726 [0.154,0.192], MRR 0.0963, a paired Δ over Markov of +0.0657 [+0.049,+0.083] (CI entirely above zero) — $\sim +61\%$ R@10, $+39\%$ MRR — on a robust plateau across $\alpha \in [0.25, 0.75]$ and reproduced across three seeds (0.1705 / 0.1726 / 0.1726). It was registered as the **seq-blend** predictor and the **blend-gru-markov** definition, the first next-track champion. Further ensembling then *failed* to improve on it (Table 2): R+A+M (0.1684) is slightly worse, R+A ($\approx$ GRU alone, 0.0894) is much worse, and both XGB stacker variants (M+R 0.1020; M+R+A 0.0887) barely clear Markov alone.

Interpretation. The win is about signal *diversity*, not model count. The GRU (latent similarity) and Markov (co-occurrence counts) rank *different* correct next-tracks, so fusing them roughly doubles the hit rate; piling on more of the *same* kind of signal — another neural latent predictor (A), redundant with R — adds nothing. The learned XGB stacker *underperforms* the simple z-blend, counter to the usual “stacking beats blending” intuition: its pointwise binary-with-negatives training does not optimize the ranking objective as well as directly fusing nor-

Table 2: The fusion campaign: the R+M z-blend beats the Markov bar decisively, and no richer ensemble improves on it. R@10 and MRR higher-better; champion bolded.

Model	R@10 [95% CI]	MRR	Note
R+M z-blend $\alpha=0.5$ (champion)	0.1726 [0.154, 0.192]	0.0963	Δ vs Markov +0.066 [+0.05, +0.08]
R+A+M z-blend	0.1684	0.0943	ANN leg redundant with GRU
B2 Markov bar (M)	0.1069 [0.092, 0.123]	0.0694	the reference
XGB stacker M+R	0.1020	0.0561	learned combiner $\ll$ z-blend
GRU alone (R)	0.0957	0.0411	ties the bar on R@10 only
R+A z-blend	0.0894	0.0377	$\approx$ GRU alone
XGB stacker M+R+A	0.0887	0.0572	barely clears Markov

malized scores. The remaining headroom is bounded by the 0.55 cold-item rate and requires *genuinely new signal*, not more model fusion — the hypothesis the next campaign tests. *Source: 2026-07-14-nexttrack-blend-and-ensemble.md.*

7 Graded-relevance metrics and new base learners: content is the only additive signal

Rationale. Two threads in one campaign. (A) Make the evaluation credit landing the right *band/vibe* even when the exact track is wrong — the observation that “the first choice is usually not the exact track, but the artist is usually right” — by promoting *artist@k*, *genre@k* and *artist_mrr* to first-class metrics wired end-to-end (`seq_common.eval_from_scores` $\rightarrow$ Rust Metrics / MetricsSpec $\rightarrow$ domain.toml $\rightarrow$ UI). (B) Try genuinely new leak-free base learners — **C** = content-kNN, **M2** = a second-order Markov trigram with back-off, and **I** = a train-only item2vec SGNS — scored on both the exact and the new graded metrics.

Results. The exact-metric path was byte-identical (purely additive keys); the champion reproduced R@10 0.1726 / MRR 0.0963 exactly. No new base learner beat the bar standalone (M2 0.1076 $\approx$ M 0.1069 — the trigram falls through to the bigram; item2vec 0.0517 is the weakest leg). Blended into the champion, **content is the only additive signal: R+M+C** (adding the content leg) is the top blend on MRR (0.0963 $\rightarrow$ 0.1014), artist@10 (0.3215 $\rightarrow$ 0.3298) and artist_mrr, and ties the champion on exact R@10 (paired Δ +0.0014, CI straddles zero). Item2vec *drags* the blend (R+M+I 0.1349); the trigram is neutral.

Interpretation. Content-kNN ranks *different* near-correct tracks (latent vibe similarity) than the GRU (session recurrence) and Markov (co-occurrence), exactly the diversity the fusion campaign predicted new signal would need. On the AE-64 space the content leg only *tied* the champion on exact recall while lifting the graded metrics, so it was carried forward as the graded-preferred third leg, not a new exact champion. That “tie” becomes a *win* once the item space is richer (Section 10): on PCA-192 the same R+M+C construction is the strongest on record. Co-listening item2vec and the trigram Markov added nothing on this data. *Source: 2026-07-14-nexttrack-graded-relevance-and-new-legs.md.*

8 The item-representation compression bake-off

Rationale. The champion’s item space — a 64-dim AE latent — was one arbitrary early choice, and the sequence GRU predicts *into* and retrieves *over* whatever geometry that space imposes. This

campaign asks which *compression* is the best space for the GRU: linear PCA versus the nonlinear autoencoder, each at latent dim 32 / 64 / 128. Six alternative spaces were baked from the *same* leak-safe 384-dim content source (`spotify_tracks_content`), each fit unsupervised over all 23 529 items using only intrinsic content/acoustic/catalog fields, then scored under a *frozen* GRU readout (`gru-infonce-h256`), so the item latent space is the only axis that moves. The design bet was that the AE’s equal-weight-per-block reconstruction loss would *preserve* the small acoustic/catalog blocks that PCA’s text-dominated global variance would starve; the null was that dim, not the compressor, sets the ceiling.

Results. The parity gate passed: the AE-64 control reproduced its on-record R@10 0.0957. All six spaces trained cleanly (Table 3; Figure 1). **PCA-128** is the single-GRU winner at R@10 0.1146, a paired Δ of +0.0189 [+0.0042, +0.0335] over the AE-64 control — a CI *entirely above zero*, the only arm that clears the control significantly — and it also tops MRR, artist@10 and artist_mrr. PCA-64 is the runner-up (0.1041, tie). The two 32-dim spaces sit far below the 64-dim control (both CIs below zero). The training-free content-kNN probe is flat at ~ 0.06 across all six spaces and does *not* track the GRU ranking.

Table 3: The item-representation compression bake-off: six item spaces (linear PCA vs the nonlinear AE $\times$ latent dim 32 / 64 / 128), each scored under the *same* frozen `gru-infonce-h256` readout (1431 test sessions; paired- Δ vs the in-batch AE-64 control). R@10, MRR, artist@10, genre@10 higher-better. Winner (PCA-128) bolded; best cell per metric bolded.

Space	Run id	R@10 [95% CI]	Δ vs AE-64	MRR	art@10	gen@10
PCA-128 (winner)	run-2026071 5-042100-a1 d9d	0.1146 [0.099 , 0.132]	+0.019 [+0.004, +0.034]	0.0489	0.3326	0.4521
PCA-64 (runner-up)	run-2026071 5-042100-88 51d	0.1041 [0.088, 0.121]	+0.008 (tie)	0.0416	0.3117	0.4598
AE-64 (control)	run-2026071 5-042059-dd 261	0.0957 [0.081, 0.111]	— (control)	0.0411	0.2942	0.4500
AE-128	run-2026071 5-042059-e9 6ae	0.0929 [0.078, 0.108]	−0.003 (tie)	0.0398	0.3026	0.4640
AE-32	run-2026071 5-042059-d3 f9e	0.0545 [0.043, 0.066]	−0.041	0.0274	0.2376	0.3955
PCA-32	run-2026071 5-042059-b1 242	0.0412 [0.031, 0.052]	−0.055	0.0224	0.2187	0.3816

The single-GRU win does **not** survive into the production blend: re-fusing the winning PCA-128 GRU leg with the Markov leg scores R@10 0.1670, *below* the AE-64 champion blend (0.1726) with overlapping CIs. Nothing was promoted; PCA-128 was carried forward only as the sanctioned Phase-2 base space.

Interpretation. Latent dimensionality is the dominant lever and the compressor is secondary. The design’s acoustic-preservation hypothesis is *refuted*: at generous dim the linear PCA beats the nonlinear AE, and although AE-32 preserves the numeric/acoustic blocks near-perfectly it ranks near the bottom. What helps is a *variance-ordered, decorrelated* target geometry that the GRU finds easier to regress into than the AE’s entangled equal-weight latents. The AE’s edge at dim 32 is a purely low-capacity phenomenon that reverses once dim is generous. The flat content-kNN probe

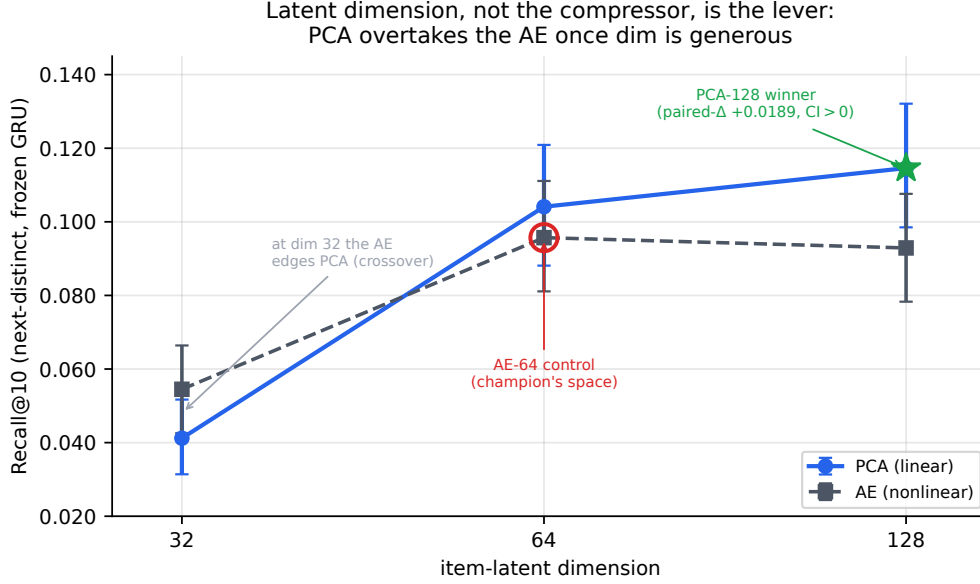


Figure 1: Latent dimension, not the compressor, is the lever. Both series climb steeply from dim 32 to 64; PCA (blue) then keeps climbing to 128 and *overtakes* the AE, while the AE (grey) plateaus — its 128-dim point actually dips below its own 64-dim control. At dim 32 the AE edges PCA, a low-capacity crossover that reverses once dim is generous. The AE-64 control (red ring) is the champion’s item space; the PCA-128 winner (green star) is the only arm whose paired- Δ CI clears zero. Error bars are bootstrap 95% CIs; the neighbouring CIs overlap, so the win rests on the *paired* test.

that fails to track the GRU ranking is a flagged divergence: the win comes from the *learnability* of PCA’s target geometry, not a demonstrably richer space. And the load-bearing caution is the blend result: a clearly stronger GRU leg bought *nothing* at the blend, because the Markov leg is representation-independent and the extra tracks the better GRU recovers overlap what Markov already supplies. *Source: 2026-07-15-nexttrack-compression-representation-bakeoff.md.*

9 The dimension ceiling: PCA peaks at 192

Rationale. With dimension established as the lever and PCA still climbing at 128, this leg extended the PCA sweep upward (PCA-192, PCA-256) to locate the real optimum before Phase 2 committed a base space, and added AE-192 to test whether $\text{PCA} \geq \text{AE}$ holds at higher dimension. The protocol was identical (same frozen `gru-infonce-h256` readout, same parity gate); the reference is PCA-128 (run-20260715-042100-a1d9d).

Results. The PCA R@10-versus-dimension curve **peaks at ~ 192** (R@10 0.1230) and *turns over* by 256 (0.1167); it was not still climbing. The full trend is $32 \rightarrow 0.041$, $64 \rightarrow 0.104$, $128 \rightarrow 0.115$, **$192 \rightarrow 0.123$** , $256 \rightarrow 0.117$ (Figure 2). PCA-192 is the point-optimum, but its paired Δ versus PCA-128 is $+0.0084$ $[-0.0007, +0.0189]$ — it misses the “CI entirely above zero” bar by 0.0007, a whisker — so **PCA-128 stands as the Phase-2 base**, with PCA-192 flagged as the confirm-and-promote candidate. $\text{PCA} \geq \text{AE}$ holds strongly and the gap is widest at 192 (PCA-192 0.123 versus AE-192 0.098); the AE plateaus flat at ~ 0.09 – 0.10 from dim 64 upward.

Interpretation. Extra variance past ~ 192 (PCA-256 captures cumulative explained-variance

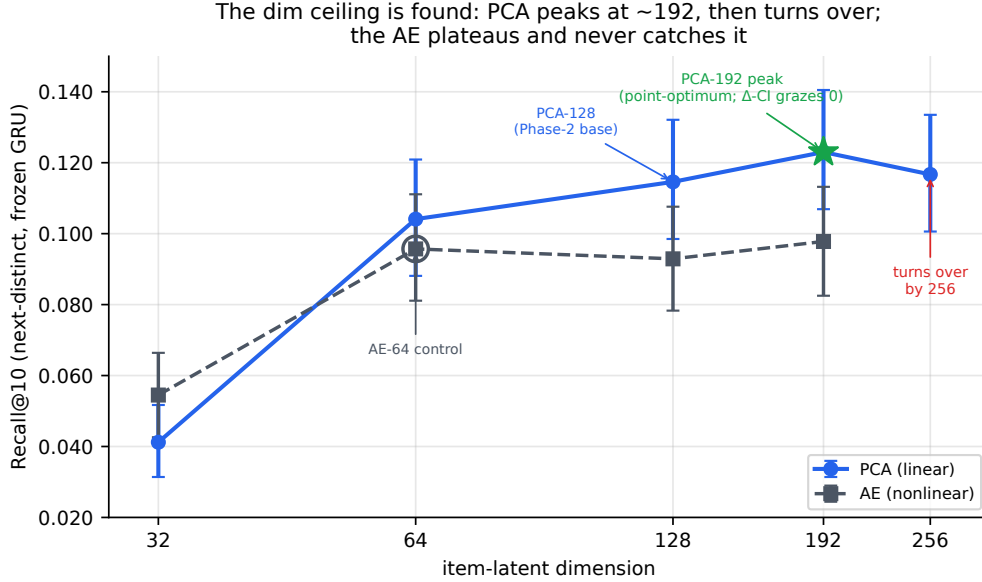


Figure 2: The dimension ceiling is found. Extending the compression sweep of Figure 1 to PCA-192/256 and AE-192, the PCA curve (blue) peaks at ~192 (R@10 0.123) and turns over by 256, while the AE (grey) plateaus flat at ~0.09–0.10 and never catches it — the $\text{PCA} \geq \text{AE}$ gap is widest at 192. PCA-192 is the point-optimum (green star) but its paired Δ over PCA-128 grazes zero, so PCA-128 stays the Phase-2 single-model base. Error bars are bootstrap 95% CIs over the 1431 test sessions.

ratio 0.997) is low-signal / noise dimensions the GRU cannot exploit and that slightly dilute the target geometry. The AE’s flat plateau reconfirms the refuted acoustic-preservation hypothesis at higher dimension: its equal-weight reconstruction objective keeps spending capacity on the tiny numeric/acoustic blocks (reconstruction $R^2 \approx 0.999$ at 192) rather than the high-variance text/catalog structure the GRU needs. As before, the training-free content-kNN probe stays flat and does not reproduce the GRU’s peak, so the dimension optimum is about target *learnability*, not intrinsic retrieval richness. *Source: 2026-07-15b-nexttrack-compression-dim-extension.md.*

10 The model-family sweep and the new champion, R+M+C on PCA-192

Rationale. With the compression axis settled (PCA-128 base, PCA-192 candidate), this campaign swept the model *family* on the chosen spaces: does a different recurrent cell (LSTM), a non-recurrent pooled MLP (ANN), or a *learned* XGB stacker beat the fixed GRU×Markov z-blend — and does adding the content-kNN third leg (R+M+C) on the *rich* PCA space finally beat the champion? A frozen held config ran across all arms; only the model family and the representation varied, and the server was serialized (`--max-runs 1`) to avoid a thread-oversubscription throughput trap.

Results. A **new best-on-record row**: the **R+M+C z-blend on PCA-192** (GRU × train-only Markov × content-kNN, equal thirds) reaches R@10 0.1859 [0.166, 0.207], MRR 0.1025, a paired Δ over the prior champion of +0.0133 [+0.0007, +0.0259] — entirely above zero, with no MRR regression (Table 4; Figure 3). The content leg is genuinely additive on the rich space,

beating same-space R+M by +0.0154 (PCA-128) / +0.0147 (PCA-192), both CIs above zero — *stronger* than the AE-64 R+M+C that only tied (Section 7), because the richer, higher-EVR space makes the content leg pay off. The learned **XGB stacker badly underperforms** the fixed z-blend (0.080–0.087, losing even to its own single GRU base leg); the **LSTM is the best single model** (PCA-128 0.1272, paired Δ over the GRU +0.0126, CI above zero) but far below the blends; and the Phase-2 single base **stays PCA-128**.

Table 4: Phase-2 model-family sweep on the rich PCA spaces, ranked by R@10 (higher-better). The fixed z-blend is the combiner of record; the learned XGB stacker underperforms it badly. Champion row and prior champion bolded; best cell per metric bolded. Paired Δ is per-session against the named reference.

Model	Run id / source	R@10 [95% CI]	MRR	paired Δ
R+M+C, PCA-192 (champ. row)	analyze_rmc (A4)	0.1859 [0.166, 0.207]	0.1025	vs champ +0.0133 [+0.001, +0.026]
R+M+C, PCA-128	analyze_rmc (A2)	0.1824 [0.162, 0.203]	0.0993	vs champ +0.0098 (tie)
R+M, AE-64 (prior champion)	run-20260714- 170928-645ae	0.1726 [0.154, 0.192]	0.0963	— (reference)
R+M, PCA-192	run-20260715- 142440-96f53	0.1712 [0.151, 0.192]	0.0952	vs champ −0.0014 (tie)
R+M, PCA-128	run-20260715- 044920-36dce	0.1670 [0.148, 0.187]	0.0920	vs champ −0.0056 (tie)
LSTM single, PCA-128	run-20260715- 142440-546d0	0.1272 [0.110, 0.145]	0.0495	vs GRU-128 +0.0126 (> 0)
GRU single, PCA-192	run-20260715- 142440-89d6a	0.1230 [0.106, 0.140]	0.0534	vs GRU-128 +0.0084 (tie)
ANN pooled-MLP, PCA-128	run-20260715- 142440-7e528	0.0901 [0.076, 0.105]	0.0394	vs GRU-128 −0.0245
XGB stacker M+G, PCA-128	run-20260715- 142440-371b1	0.0867 [0.073, 0.101]	0.0493	vs GRU-128 −0.0280
XGB stacker M+G+A, PCA-192	run-20260715- 142440-f111d	0.0853 [0.071, 0.099]	0.0490	vs GRU-192 −0.0377

The crown, and its two honest caveats. The board-event candidate was confirmed by reproducing the GRU-192 leg five times (the original plus four reruns) and recomputing R+M+C each time; the paired Δ over the prior champion was +0.0133 [+0.0007, +0.0259] *every* rerun, with the MRR lead intact, so R+M+C on PCA-192 was recorded as the new champion row. Two caveats travel with the crown and must not be omitted. First, **the reproduction spread was exactly zero** because **seq-nexttrack** training is deterministic on this CPU — all five GRU-192 checkpoints were byte-identical — so the confirm verified *determinism, not seed-robustness*; the only uncertainty quantified is the single-split bootstrap. Second, **the paired Δ lower bound is +0.0007**: the crown clears zero by a whisker on a single split. The crown is real by the pre-agreed criterion, but the margin over the prior champion is thin, and hardening it needs a second leak-free test split (or seed-varied legs once the blend is generalized to three legs).

Interpretation. The compression-plus-model-family thread is otherwise closed: PCA-128 is the single-model base, the LSTM is the best single model, the learned stacker is dead as a combiner, and the fixed z-blend is the combiner of record. The load-bearing result is that *content is the one leg that adds genuinely new ranking signal, and it pays off more on the higher-EVR PCA space* — the exception to the standing rule that single-model / representation gains do not carry to the blend (the R+M pair on the rich PCA spaces only ties the AE-64 champion). The productization

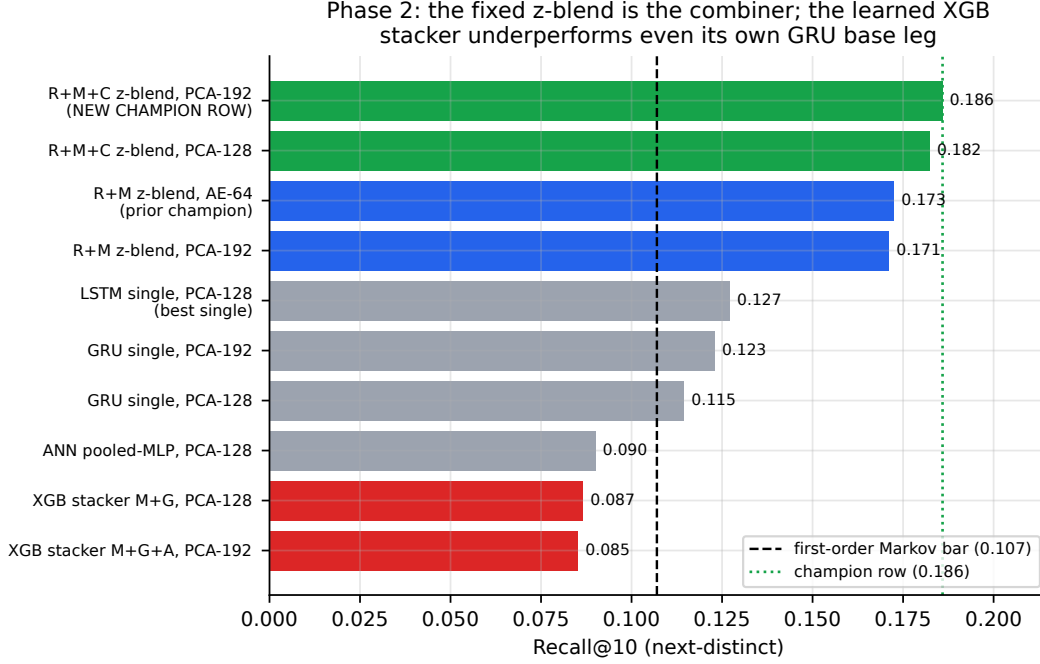


Figure 3: The Phase-2 model-family sweep. The fixed z-blend is the combiner of record: R+M+C on PCA-192 (green) is the new champion row, the content leg is genuinely additive on the rich space, and the learned XGB stacker (red) badly underperforms — it loses even to its own single GRU base leg. LSTM edges GRU as the best single model (grey) but sits far below the blends. The first-order Markov bar (0.107) is the incumbent reference; the champion row is at 0.186.

gap remains: the champion is train-only and off-server, so the prior champion R+M on AE-64 stays the deployable anchor until the three-leg blend is registered with a `predict` subcommand.

Source: `2026-07-15c-nexttrack-phase2-model-sweep.md`.

11 Conclusions

Seven campaigns (Jul 2026) on the next-track retrieval task converge on a clear story. (1) A session sequence model, even at its best (GRU-infonce), only *ties* the first-order Markov co-listening bar on Recall@10 and loses on MRR: predicting a next-item latent and retrieving by one cosine hop is a harder route to the exact next track than a bigram that directly memorizes adjacency. (2) The training *loss* is the dominant single-model lever (InfoNCE $\approx 2.3\times$ cosine); network *topology* — depth, width, residual, direction — is a spent lever, and no architecture beats the bar. (3) The breakthrough is *rank fusion*: an R+M z-blend of the GRU and Markov legs beats the bar by $\sim +61\%$ Recall@10 (three-seed confirmed), because the two legs rank *different* correct tracks. The win is about signal *diversity*, not model count — adding a redundant neural leg does not help, and a learned XGB stacker *underperforms* the fixed z-blend, because its pointwise training does not capture the ranking objective. (4) An item-representation sweep found latent *dimension*, not the compressor, to be the lever: linear PCA beats the nonlinear autoencoder at every $\dim \geq 64$ (the acoustic-preservation hypothesis is refuted), the curve peaks at PCA-192 and turns over by 256, and PCA-128 is the confirmed single-model base. (5) On the rich PCA space the one genuinely additive new signal — *content-kNN*, which only *tied* the champion on the AE-64 space — becomes

a win: a three-leg R+M+C z-blend on PCA-192 is the current best-on-record row at Recall@10 0.1859, MRR 0.1025 (paired Δ over the prior champion +0.0133, CI above zero, confirmed across five reproductions). (6) Two honest caveats bound the crown: the reproduction confirm had zero spread because `seq-nexttrack` is deterministic on this CPU (so it verified determinism, not seed-robustness), and the paired- Δ lower bound clears zero by only +0.0007 on a single split — hardening it needs a second leak-free test split. (7) Every ranking predictor here is train-only and off-server, so the crown is a best-on-record *row*, not a promoted, app-exportable model; the prior champion R+M on AE-64 remains the productization anchor, and registering a three-leg blend with a `predict` subcommand is the standing top follow-up.

12 Source reports

This synthesis consolidates seven next-track campaign reports from `experiments/`:

- `2026-07-13-nexttrack-sequence-sweep.md` — the founding next-track campaign: a GRU/LSTM sequence sweep against the first-order Markov bar; the best GRU-infonce ties on R@10 but loses on MRR (Section 4).
- `2026-07-13-nexttrack-topology-scan.md` — the depth / width / residual / bidirectional topology scan; architecture is a spent lever, all arms cap below the Markov bar (Section 5).
- `2026-07-14-nexttrack-blend-and-ensemble.md` — the GRU $\times$ Markov R+M z-blend beats the bar +61% R@10 (the first next-track champion); no richer ensemble improves on it and the learned XGB stacker underperforms (Section 6).
- `2026-07-14-nexttrack-graded-relevance-and-new-legs.md` — first-class graded-relevance metrics (artist@k / genre@k / artist_mrr) and new leak-free base learners; content-kNN is the only additive signal, tying the champion on AE-64 (Section 7).
- `2026-07-15-nexttrack-compression-representation-bakeoff.md` — the item-representation compression bake-off (PCA vs autoencoder $\times$ latent dim 32/64/128) under a frozen GRU readout; PCA-128 is the single-GRU winner but the win does not carry into the blend (Section 8).
- `2026-07-15b-nexttrack-compression-dim-extension.md` — the PCA/AE dimension sweep extended to 192/256; the curve peaks at PCA-192 and turns over, so PCA-128 stays the Phase-2 base (Section 9).
- `2026-07-15c-nexttrack-phase2-model-sweep.md` — the model-family sweep on the rich PCA spaces; the fixed z-blend beats the learned stacker, LSTM is the best single model, and R+M+C on PCA-192 is crowned the new best-on-record row (Section 10).